

Unit 3 - Understanding Rational Numbers

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The following B.E.S.T. Standards will be covered in this unit:

MA.6.NSO.1.1 Extend previous understanding of numbers to define rational numbers. Plot, order, and compare rational numbers.

- *Clarification 1: Within this benchmark, the expectation is to plot, order, and compare positive and negative rational numbers when given in the same form and to plot, order, and compare positive rational numbers when given in different forms (fraction, decimal, percentage).*
- *Clarification 2: Within this benchmark, the expectation is to use symbols ($<$, $>$, or $=$).*

MA.6.NSO.1.2 Given a mathematical or real-world context, represent quantities that have opposite direction using rational numbers. Compare them on a number line and explain the meaning of zero within its context.

- *Clarification 1: Instruction includes vertical and horizontal number lines, context referring to distances, temperatures and finances, and using informal verbal comparisons, such as, lower, warmer, or more in debt.*
- *Clarification 2: Within this benchmark, the expectation is to compare positive and negative rational numbers when given in the same form.*

MA.6.NSO.1.3 Given a mathematical or real-world context, interpret the absolute value of a number as the distance from zero on a number line. Find the absolute value of rational numbers.

- *Clarification 1: Instruction includes the connection of absolute value to mirror images about zero and to opposites.*
- *Clarification 2: Instruction includes vertical and horizontal number lines and context referring to distances, temperature, and finances.*

MA.6.NSO.1.4 Solve mathematical and real-world problems involving absolute value, including the comparison of absolute value.

- *Clarification 1: Absolute value situations include distances, temperatures, and finances.*
- *Clarification 2: Problems involving calculations with absolute value are limited to two or fewer operations.*
- *Clarification 3: Within this benchmark, the expectation is to use integers only.*

Unit 3, Lesson 1: Rational Numbers and the Number Line – Part 1



Benchmark

MA.6.NSO.1.2 Given a mathematical or real-world context, represent quantities that have opposite direction using rational numbers. Compare them on a number line and explain the meaning of zero within its context.



Learning Targets

- ☐ I can demonstrate an understanding of negative numbers.
- ☐ I can recognize that a number in the real number system can have a variety of classifications.
- ☐ I can plot rational numbers on a horizontal and vertical number line.



3.1.1 Warm-Up: Creating Zero

1. Use only digits 1 to 9, at most one time each, so that the equation is true.

$$(\square + \square - \square + \square) - (\square + \square - \square + \square) = 0$$

2. Switch your answer with a partner to check. Ask your partner his/her strategy for finding his/her answer and write his/her strategy below.



3.1.2 Exploration: How Can Negative Numbers Exist?

Negative numbers are numbers less than zero.

Examples include:

- a temperature below 0 degrees
- the amount of money owed, or debt
- a contestant's score on Jeopardy (when someone misses a question, he/she loses points, so his/her score could be a negative amount)

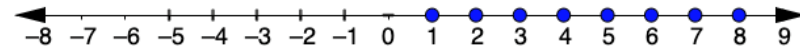
1. Discuss the examples of negative numbers with your partner.
2. Discuss with your partner which of the contexts in the table could have a value less than 0, then write an example or explain your thinking.

Context	Example or Explanation
Direction	
Points in a game	
Time	



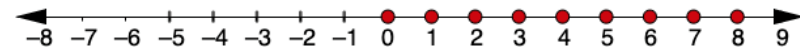
3.1.3 Guided Instruction: The Real Number System

The values in blue on the number line are some of the numbers in the set of **natural numbers**.



Natural numbers are the counting numbers {1, 2, 3, 4, 5 ...}.

The values in red on the number line are some of the numbers in the set of **whole numbers**.



Whole numbers are natural numbers and zero. {0, 1, 2, 3, 4, 5 ...}

The values in green on the number line are some of the values in the set of **integers**.



Integers are whole numbers and their opposites. {... -4, -3, -2, -1, 0, 1, 2, 3, 4 ...}

1. Use the number lines to help to answer each of the following questions.

- What do you notice about the set of natural numbers and the set of whole numbers?
- What do you notice about the set of integers?
- What type of number is -10 ?
- What type of number is 27 ?
- Describe the numbers that appear in all three sets.

Camaria wondered to herself, "What about numbers such as $3\frac{1}{2}$ and -1.75 ?"

These numbers are examples of **rational numbers**.



Rational numbers are numbers that can be expressed as the ratio of two integers.

2. Use the definition of rational numbers to write each as a ratio of two integers.

Number	Ratio of Two Integers
-1.5	
9	

In your study of mathematics, you will learn about **irrational numbers**, which are numbers that cannot be written as a ratio of two integers.



The set of **real numbers** is the set of all rational and irrational numbers.



3.1.4 Activity: Horizontal Number Line

Using a sheet of paper, follow the steps to make your own number line.

Step 1: Use a straightedge or a ruler to draw a horizontal line. Mark the middle point of the line and label it 0.

Step 2: To the right of 0, draw tick marks that are 1 centimeter apart. Label the tick marks 1, 2, 3 ... 10.

Step 3: Fold your paper so that a vertical crease goes through 0 and the two sides of the number line match up perfectly.

Step 4: Use the fold to help you trace the tick marks that you already drew onto the opposite side of the number line. Unfold and label the tick marks $-1, -2, -3 \dots -10$. This represents the negative side of your number line.

Use the number line to answer each of the following questions.

1. Describe how far away the number 5 is from 0.

The number 5 is _____ units to the _____ of 0 on the number line.

2. Write a positive number and a negative number where each is further away from 0 than the number 5.

3. Describe how far away the number -2 is from 0.

The number -2 is _____ units to the _____ of 0 on the number line.

4. Write a positive number and a negative number that are each further away from 0 than -2 .



3.1.5 Bring It Together: Plotting Rational Numbers on Number Lines

When plotting numbers on a number line, the sign of the number indicates which direction from 0 to move on the number line.

- On a horizontal number line, positive numbers are to the

☐ right

☐ left

 of zero.
- On a vertical number line, positive numbers are

☐ above

☐ below

 zero.
- On a horizontal number line, negative numbers are to the

☐ right

☐ left

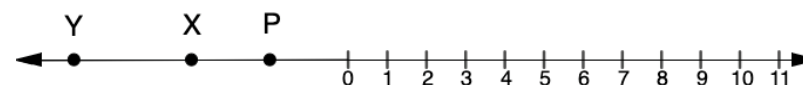
 of zero.
- On a vertical number line, negative numbers are

☐ above

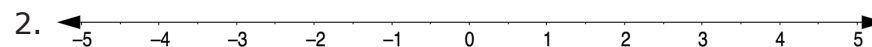
☐ below

 zero.

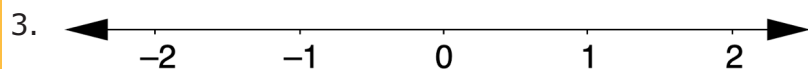
1. A number line is shown with points P , X , and Y .



- What type of numbers are P , X , and Y ?
- Estimate the location of points P , X , and Y .



- Plot the values $-3\frac{2}{3}$ and 4.3 on the number line.
- How did the tick marks between each integer help you determine where to plot 4.3 ?
- How did the tick marks between each integer help you determine where to plot $-3\frac{2}{3}$?

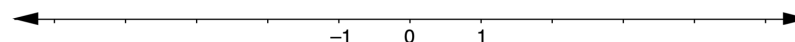


- On the number line, make a tick mark that is halfway between each value on the number line. Label each tick mark with the value of its location.
- Make another tick mark halfway between each value and the tick marks that were previously made. Label each with the value of its location.
- Plot the values $-1\frac{7}{12}$ and 0.25 on the number line.

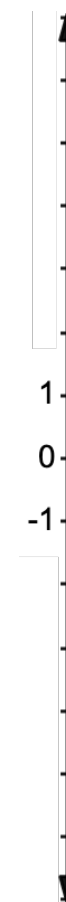


3.1.6 Try It: Plotting Rational Numbers on a Number Line

- Plot the numbers $-\frac{5}{2}$, -4 , $\frac{1}{4}$, and 3 on the horizontal number line.



- Plot the rational numbers -0.25 , -1.8 , 6.2 , and 4.75 on the vertical number line.

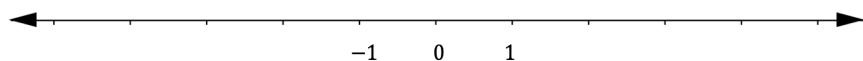




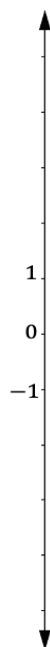
3.1.7 Your Turn!

- Negative numbers are ☐ greater ☐ less than zero. On a horizontal number line, negative numbers are to the ☐ right ☐ left of zero. On a vertical number line, negative numbers are ☐ above ☐ below zero.

- Plot the integers 2, -3, 0, -5 and 5 on the horizontal number line.

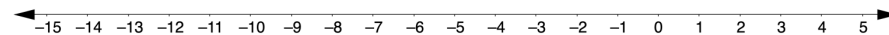


- Plot the rational numbers $-1\frac{1}{5}$, $-\frac{4}{5}$, 2, and $\frac{6}{5}$ on the vertical number line.

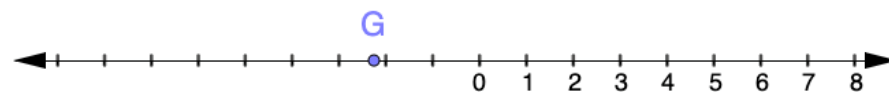


3.1.8 Wrap-Up: Ticket Out the Door

- Plot the rational numbers $\frac{-25}{4}$, $-12\frac{3}{4}$, $\frac{12}{4}$, and $\frac{1}{4}$ on the number line.



- Explain how to estimate the value of G .



Unit 3, Lesson 2: Rational Numbers and the Number Line – Part 2



Benchmark

MA.6.NSO.1.2 Given a mathematical or real-world context, represent quantities that have opposite direction using rational numbers. Compare them on a number line and explain the meaning of zero within its context.



Learning Targets

- ☐ I can determine how far a number and its opposite are from zero.
- ☐ I can represent a number and its opposite on a number line.
- ☐ I can compare a quantity and its opposite on a horizontal or vertical line and explain the meaning of zero within the context.



3.2.1 Warm-Up: Word Pairs

Up	Down
Left	Right
Clockwise	Counterclockwise
Increase	Decrease
Ebb	Flow
West	East
North	South
Gain	Lose

1. What do you notice about the pairs of words above?

2. How do you think this may be used in the study of mathematics?



3.2.2 Activity: Folded Number Line

On a separate sheet of paper, create a number line with integers from -10 to 10 to answer the following questions. Fold the number line so that the vertical crease goes through 0 and the two sides of the number line match up perfectly.

1. When the number line is folded, which number is the same distance away from 0 as the number 4 ?
2. When the number line is folded, which number is the same distance away from 0 as the number -7 ?
3. When the number line is folded, what do you notice about the numbers that are the same distance away from 0 ?
4. Find another pair of numbers that are the same distance away from 0 .
5. Two numbers that are the same distance from zero on the number line are called _____.

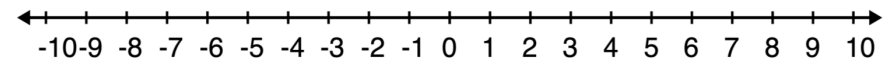


3.2.3 Collaboration: Opposites

With your partner, choose values for W and F . The value for W can be any rational number between -8 and 0 that is not an integer. The value for F can be any rational number between 0 and 8 that is not an integer.

The value for W is _____. The value for F is _____.

1. Plot and label both numbers on the number line.



2. Plot the opposite of W . Label the point X .
3. What is the value of point X ?
4. How far is point X from 0 ?
5. Plot the opposite of F . Label the point G .
6. What is the value of point G ?
7. How far is point G from 0 ?

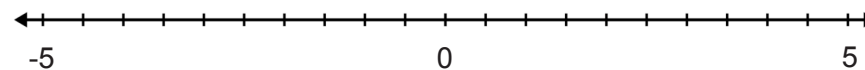
8. Noah said, "The expression $-a$ can be read 'negative a ' or 'the opposite of a '. So, if a is a rational number, $-a$ will always be a negative number."

Use points F , G , W , and X to explain to Noah that the opposite of a value could be positive.



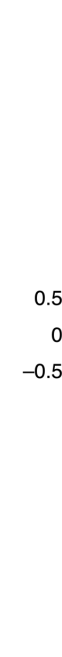
3.2.4 Try It: Opposites

1. Use the number line below to answer the following problems.



- Label the tick marks on the number line. Then plot the value $\frac{5}{4}$ and its opposite on the number line.
- Which value is larger, $\frac{5}{4}$ or its opposite?

2. Use the vertical number line to answer the following problems.



- Label the tick marks on the number line. Plot and label the value -3.8 and its opposite on the number line.
- Which value is larger, -3.8 or its opposite?



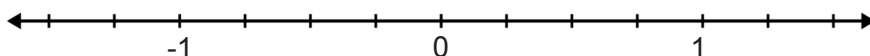


3.2.5 Guided Instruction: Comparing a Quantity and Its Opposite

Plot each pair of values on the number line. Choose an answer to correctly fill in the blank.

1.

- a. Plot -0.25 and 0.25 .

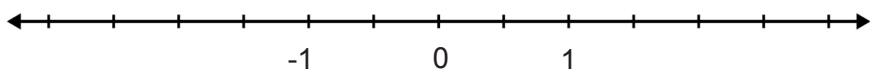


- b. The value -0.25 is ☐ greater ☐ less than 0.25 .

- c. -0.25 ☐ $>$ ☐ $<$ 0.25

2.

- a. Plot $2\frac{2}{5}$ and $-2\frac{2}{5}$.



- b. The value $2\frac{2}{5}$ is ☐ greater ☐ less than $-2\frac{2}{5}$.

- c. $2\frac{2}{5}$ ☐ $>$ ☐ $<$ $-2\frac{2}{5}$

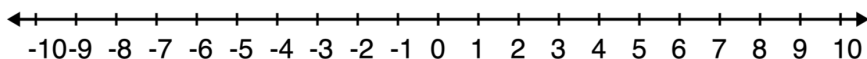
3. In each comparison, the number on a horizontal number line that is farther to the ☐ right ☐ left is the greater value.

4. When comparing opposites, the ☐ positive ☐ negative value will always be the greater value.



3.2.6 Your Turn!

- Plot and label each value and its opposite on the number line: 9.5, -2 , 4.2 , $-7\frac{3}{4}$.



- Pairs of opposite numbers are
☐ the same
☐ a different
 distance from zero on a number line.

- Complete each comparison statement for the pairs of numbers you plotted in problem #1. For each statement, use $<$ or $>$ in the first blank and the opposite of the given number in the second blank.

a. 9.5 _____

b. -2 _____

c. 4.2 _____

d. $-7\frac{3}{4}$ _____



3.2.7 Wrap-Up: Cool Down

- What is the relationship between a number and its opposite? Use the words **number line** and **zero** in your explanation.

Unit 3, Lesson 3: Plotting, Ordering, and Comparing Rational Numbers – Part 1



Benchmark

MA.6.NSO.1.1 Extend previous understanding of numbers to define rational numbers. Plot, order, and compare rational numbers.



Learning Targets

- ☐ I can plot and compare rational numbers represented as a decimal or percent.
- ☐ I can plot, compare, and order fractional representations of rational numbers.



3.3.1 Warm-Up: Notice and Wonder

- The diagram gives the temperature, in Fahrenheit ($^{\circ}\text{F}$) and in Celsius ($^{\circ}\text{C}$), for Memphis, Tennessee, and Bangor, Maine.



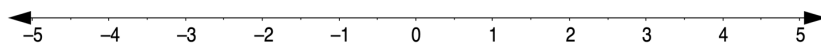
What do you notice about the temperatures?



3.3.2 Guided Instruction: Comparing and Ordering Decimals

1. The table shows the lowest temperature, in degrees Celsius ($^{\circ}\text{C}$), for each day during the week of October 14, 2019, in Anchorage, Alaska.

Day	Mon	Tue	Wed	Thu	Fri	Sat	Sun
Temperature ($^{\circ}\text{C}$)	4.2	-3.7	-0.75	1.3	2.8	-0.1	-2.6



- Plot the temperatures on the number line.
- Which temperature did you plot farthest to the left?
- What day of the week is represented by the value that was plotted farthest to the left?

- d. Why does the location of the temperature that is plotted farthest to the left help determine which day had the lowermost temperature?

- e. Interpret the temperature in this context.

During the week of October 14, 2019, the lowermost temperature was _____ $^{\circ}\text{C}$. This was the

- ☐ warmest
- ☐ coldest

- f. The inequality symbols, $<$ or $>$, can be used to compare two of the daily low temperatures. Write an inequality that shows the relationship between two of the low temperatures for the week of October 14, 2019.

2. Jayden, a data analyst, is comparing the populations of four different states to the population of the United States. His data is shown in the table.

State	Population as a Portion of the U.S. Population (decimal form)	Population as a Percentage of the U.S. Population (percent form)
California	0.119	
Florida		6.4%
New York		5.9%
Texas	0.087	

- a. Choose two states to compare. Write or describe your comparison.

- b. Olivia and Anthony's strategies for comparing California and New York are explained below.

Olivia explained, "I know that percent represents the part per 100, so the population of New York is approximately $\frac{6}{100}$ of the population of the U.S. The portion of the U.S. population living in California is approximately 12 hundredths, or $\frac{12}{100}$. Therefore, more people live in California than in New York."

Anthony explained, "I can convert 5.9% to a decimal by dividing 5.9 by 100. So 5.9% can also be represented by 0.059. I know 0.119 is greater than 0.059. Therefore, more people live in California than in New York."

Olivia and Anthony both used different methods to come to the same conclusion. Discuss their methods with a partner.

Olivia's Method	Anthony's Method



- c. Complete each inequality statement below to compare the given values.

0.119 ____ 6.4% 6.4% ____ 0.087 0.087 ____ 5.9%

- d. In which of the four states does the highest portion of the U.S. population live?

- e. In which of the four states does the smallest portion of the U.S. population live?



3.3.3 Exploration: Plotting and Comparing Fractions

1. Plot and label the rational numbers on the number line below.

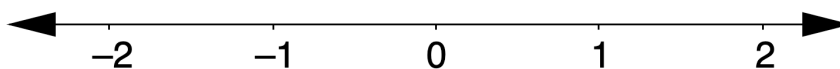
$$\frac{-3}{2}$$

$$\frac{5}{8}$$

$$\frac{5}{4}$$

$$\frac{1}{4}$$

$$\frac{-3}{4}$$



Using the rational numbers plotted above, answer the following questions.

2. Andre says that $\frac{1}{4}$ is less than $-\frac{3}{4}$ because out of the two values, $\frac{1}{4}$ is closer to 0. Explain why you agree or disagree with Andre.

3. Complete the table for each pair of numbers.

Pair	Farthest Distance From 0	Greater Value
$\frac{1}{4}$ or $\frac{5}{4}$		
$-\frac{3}{4}$ or $\frac{5}{8}$		

4. Is the value that is farther from 0 always the greater value? Explain your reasoning.



3.3.4 Your Turn!

1. Explain why each inequality statement is true or false.

a. $-5.2 < -7.1$

b. $3.1 > 38\%$

2. Use the symbols $<$, $>$ or $=$ to compare each pair of values.

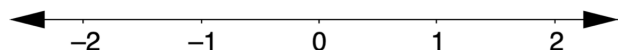
a. $-\frac{4}{3}$ _____ $-\frac{3}{2}$

b. 0.75 _____ $\frac{3}{2}$

c. $\frac{3}{2}$ _____ 1.5

3. Plot the following values on the number line. Label each point with its numeric value.

$$1.5 \quad -\frac{3}{2} \quad -\frac{4}{3} \quad 0.75 \quad \frac{3}{2}$$



- a. Complete the inequality statement to show the relationship between the values.

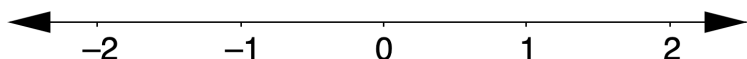
_____ > _____ > _____ > _____

4. Select all the values that are greater than -5 .

- ☐ 1.324
- ☐ -6.2
- ☐ -12.67
- ☐ 0.231
- ☐ -1.4
- ☐ -4.789

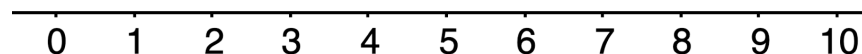
5. Order these values from least to greatest.

$$0.43, \quad -4\frac{1}{5}, \quad 0, \quad 50\%, \quad -\frac{3}{2}, \quad -6.712, \quad 19\%$$



3.3.5 Wrap-Up: Reflection

1. Circle the emoji that best represents your understanding of the lesson today.



2. Write an explanation of why you chose that emoji.

Unit 3, Lesson 4: Plotting, Ordering, and Comparing Rational Numbers – Part 2



Benchmark

MA.6.NSO.1.1 Extend previous understanding of numbers to define rational numbers. Plot, order, and compare rational numbers.



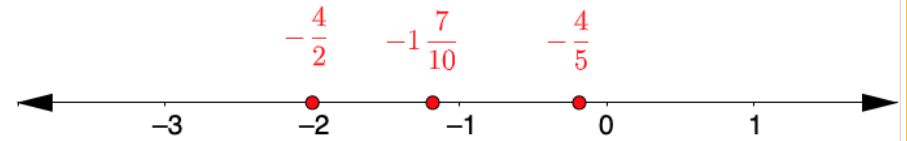
Learning Target

- ☐ I can plot, compare, and order rational numbers.



3.4.1 Warm-Up: Find the Error

A student plotted the following numbers.



1. Which values are plotted correctly?
2. Which values are NOT plotted correctly?



3.4.2 Activity: Class Number Line

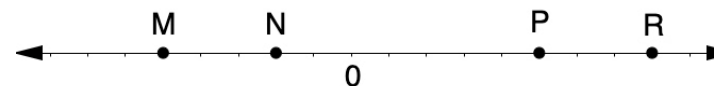
Your teacher will give you a sticky note with a rational number written on the front.

Write your name on the back of the sticky note and place your number in the correct location on the class number line.



3.4.3 Exploration: Comparing Points on a Number Line

A number line is shown with points M , N , P , and R . Work with your partner to answer the questions about the number line.



1. Use each of the given phrases to describe or compare the values of points M , N , P , R . Each statement will use one or two of the points.

Phrase	Description or Comparison of M , N , P or R
less than	
opposite of	
greater than	
negative number	

2. If P is $2\frac{1}{2}$, what are the values of M , N , and R ?
3. If N is -0.4 , what are the values of M , P , and R ?

4. If R is 200%, what are the values of M, N , and P ?

5. If M is $-1\frac{2}{3}$, what are the values of N, P , and R ?



3.4.4 Guided Instruction: Comparing Rational Numbers in Different Forms

1. Use the symbols $>$, $<$, or $=$ to compare each pair of values.

$$9.02 \text{ ____ } 9.2\% \qquad -\frac{3}{5} \text{ ____ } -\frac{13}{20} \qquad 6.050 \text{ ____ } 6.05$$

$$0.4 \text{ ____ } \frac{9}{40} \qquad 2\frac{3}{4} \text{ ____ } 2.624 \qquad -4\frac{1}{3} \text{ ____ } -\frac{10}{3}$$

2. Emma and Noah shared their strategy for comparing values written as fractions. Write your strategy in the box and note any **differences** from Emma's or Noah's strategy.

Emma's Strategy	Noah's Strategy
When Emma compares a negative fraction to a negative fraction, she starts by thinking about which whole numbers each fraction lies between and which each is closest to.	When Noah compares a negative fraction to a negative fraction, he uses common denominators.

3. Angel and Sophia share their strategies for comparing a value written as a decimal to a value written as a fraction. Write your strategy in the box and note any **similarities** from Angel's or Sophia's strategy.

Angel's Strategy	Sophia's Strategy
When Angel compares a positive decimal to a positive fraction, he rewrites the fraction as an equivalent decimal.	When Sophia compares a positive decimal to a positive fraction, she plots them on a number line.

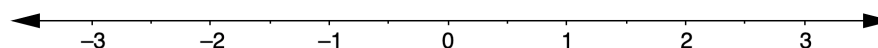


3.4.5 Try It: Plotting, Ordering, and Comparing Rational Numbers in Different Forms

1.

- a. Plot each value on the number line. Label each point with its numeric value.

-2.7 $\frac{7}{10}$ $-1\frac{1}{8}$ 0.4 $-\frac{5}{2}$ 28%



- b. What strategies did you use that were helpful when plotting the rational numbers written in different forms on the number line?

- c. Order the values from greatest to least.

- d. Complete the inequality statement to show the relationship between the values plotted on the number line.

_____ < _____ < _____ < _____ < _____ < _____



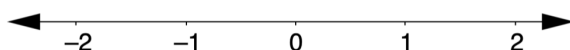
3.4.6 Your Turn!

1. Order the values from least to greatest.

$$\frac{1}{2} \quad 0 \quad 1 \quad -1\frac{1}{2} \quad -\frac{1}{2} \quad -1$$

2. Plot the values on the number line. Label each point with its numeric value.

$$0.4 \quad -1\frac{7}{10} \quad -\frac{11}{10}$$



3. Order the values from greatest to least.

$$-2 \quad 0 \quad -\frac{1}{6} \quad 1\frac{3}{8} \quad -3\frac{1}{3} \quad 1.4 \quad \frac{5}{4} \quad -\frac{13}{10} \quad \frac{21}{5}$$

4. Use the symbols $>$, $<$, or $=$ to compare each pair of values.

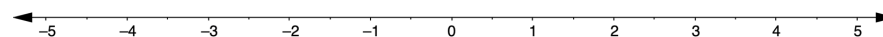
$$8.7 \text{ ____ } 8.689 \quad -\frac{5}{3} \text{ ____ } -\frac{7}{10} \quad 7.2 \text{ ____ } 7\frac{1}{3}$$

$$0.38 \text{ ____ } 5\% \quad -4\frac{5}{8} \text{ ____ } -4.5 \quad -3\frac{2}{5} \text{ ____ } -3.4$$



3.4.7 Wrap-Up: Ticket Out the Door

1. Use the values 3.6 , $-4\frac{4}{5}$, 0.7 , $-1\frac{1}{5}$, $\frac{15}{4}$, 89% , and $-\frac{7}{8}$. Plot the values on the number line. Label each point with its numeric value.



2. List the values 3.6 , $-4\frac{4}{5}$, 0.7 , $-1\frac{1}{5}$, $\frac{15}{4}$, 89% , and $-\frac{7}{8}$ from least to greatest.

3. Use the symbols $<$, $>$, or $=$ to compare each pair of values.

$$-12.742 \text{ ____ } -12.9 \quad 1\frac{2}{9} \text{ ____ } 150\% \quad 3\frac{7}{12} \text{ ____ } 3.7$$

Unit 3, Lesson 5: Rational Numbers in Real-World Context



Benchmark

MA.6.NSO.1.2 Given a mathematical or real-world context, represent quantities that have opposite directions using rational numbers. Compare them on a number line and explain the meaning of zero within its context.



Learning Targets

- ☐ I can determine positive and negative values in relation to real-world contexts.
- ☐ I can plot negative rational values given a real-world context.



3.5.1 Warm-Up: Mathematician Katherine Johnson

Born in 1918, Katherine Johnson was always fascinated by numbers. Graduating from high school at 14 and from college



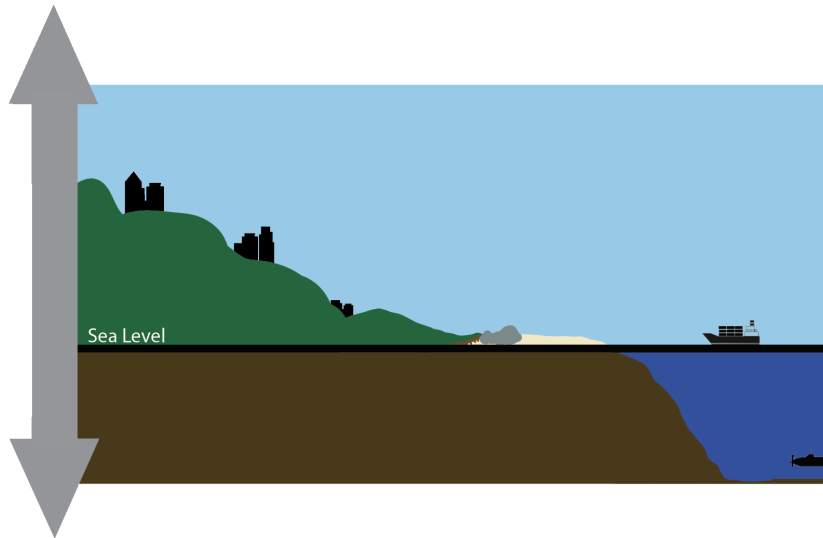
at 18, she went on to become one of the first African American mathematicians with the job title of “computer” at NASA, where she worked for three decades. As a computer, she personally checked the calculations made by electronic computers. Katherine calculated the launch window for America’s first human space flight. Her calculations were critical to the success of the Apollo Moon landing program and the start of NASA’s Space Shuttle program. In 2015, she received the Presidential Medal of Freedom from President Obama. In its official announcement, the White House cited Katherine’s influence on every major space program from Mercury to the Shuttle Program.

1. What stands out to you the most about Katherine Johnson?



3.5.2 Collaboration: Exploring Elevation

The picture shows different locations relative to sea level.



1. Describe how the picture can be used to explain the word *elevation*.

2. Ask two classmates for their description of the word elevation. Write their description in the table. *You are the only person who should write in the first column of the table.* Have your classmate initial next to your summary of their description stating that your summary is correct.

My Summary of _____'s Description of Elevation	Initials
My Summary of _____'s Description of Elevation	Initials

3. Using the picture, name one thing that is below sea level.
4. Name one thing that is above sea level.

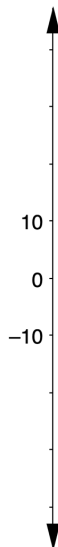


3.5.3 Guided Instruction: Positive and Negative Values in the Real World

Elevation is the vertical distance a location or object is from a given level, such as sea level.

Elevation is represented using positive and negative values.

1. If sea level is equivalent to an elevation of 0 units, then what type of value is an elevation that is below sea level?
2. Write each situation as an integer. Plot it on the vertical number line.
 - a. A hiker is 37 feet above sea level.
 - b. A diver is 19 feet below sea level.
 - c. How are these two integers related?
 - d. What does 0 represent in this situation?



3. Samuel L. Gravely, Jr. pioneered the way with a plethora of firsts for African Americans in the military. Gravely was the first African American Navy Vice Admiral, first African American to command a Navy warship, first African American to command a warship during combat, first African American to command a Navy fleet, and first African American to obtain Flag Rank in the military. If the Navy fleet included a submarine that was submerged 780 feet below sea level, which students' interpretation of the situation is correct? Circle the name of the student with the correct answer. For any statement that has an error, write a corrected statement.

Tianna	Keagan
The depth of the submarine is -780 feet below sea level.	780 feet below sea level can be represented by the integer -780 .

A bank **deposit** is the amount of money a person gives the bank for safekeeping in their account.

A bank **withdrawal** is the amount of money removed from a bank account.

An **expense** is a cost or the amount of money spent.

4. The following people have money in their bank accounts. Determine the positive and negative values that represent the effect each situation has on their account balance.

Real-World Context	Value
Paul paid \$190.90 in taxes.	
Marie made a deposit of \$50.	
Tyler made a withdrawal of \$200.	
Hannah gets paid an income of \$500 per week that she deposits in her account.	
Kaden pays \$758.91 in monthly expenses.	



3.5.4 Collaboration: Above or Below Zero

1. Represent each change in temperature on the number line. Answer the question for each situation.

At noon, the temperature was 5 degrees Celsius. By late afternoon, the temperature had risen by 6 degrees Celsius.		What was the temperature late in the afternoon?
The temperature was 8 degrees Celsius at midnight. By dawn, the temperature had dropped by 12 degrees Celsius.		What was the temperature at dawn?
Water freezes at 0 degrees Celsius, but the freezing temperature can be lowered by adding salt to the water. A student discovered that adding half a cup of salt to a gallon of water lowers its freezing temperature by 7 degrees Celsius.		What is the freezing temperature of the gallon of salt water?

Numbers below 0 are useful in contexts other than temperature.

2. With your partner, discuss and list other examples where numbers below zero can be used to represent a situation.



3.5.5 Try It: Representing the Real World With Positive and Negative Values

1. In a bank account, every time money is added or taken out of the account, a new balance is calculated. What does each of the following mean?
 - a. an account balance of $-\$23.56$
 - b. an account balance of $\$143.21$
 - c. an account balance of $\$0$

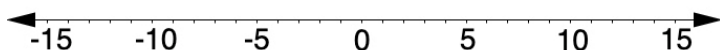
2. The change in position of the ball during each play of a football game is measured in yards.

a. What integer best represents a gain of 5 yards?

b. What integer best represents a loss of 15 yards?

c. What does the number 0 represent in this context?

d. Plot all three values on the number line.

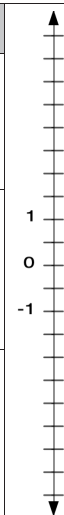


3. The average monthly rainfall for a city was calculated based on records kept over the past 50 years. The chart shows the difference between the average monthly rainfall and the rainfall amount this year for these months.

Month	Difference From Monthly Average (in inches)
January	$-2\frac{3}{4}$
February	$1\frac{5}{8}$
March	0
April	$-\frac{3}{8}$
May	$4\frac{1}{4}$

Explain what the value in the chart means for the following months in the context of this problem. Then, plot and label each month's value on the number line.

Month	Interpretation
February	
March	
April	





3.5.6 Your Turn!

1. Write a value that represents each description.

- a. an expense of \$40.23 _____
- b. a deposit of \$225.89 _____
- c. 16,000 feet above sea level _____
- d. a temperature increase of 14°F _____
- e. a withdrawal of \$271.49 _____
- f. a gain of 17 yards _____

2. Read each statement and select the correct description of each situation.

a. A diver is 60 feet below the surface of the ocean.

- ☐ The depth of the diver is 60 feet below the ocean's surface.
- ☐ The diver is -60 feet below the ocean's surface.

b. The elevation of the bottom of a boat with respect to sea level is given as -5 feet.

- ☐ The bottom of the boat is 5 feet above sea level.
- ☐ The bottom of the boat is 5 feet below sea level.



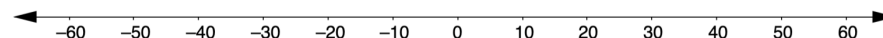
3.5.7 Wrap-Up: Ticket Out the Door

Keisha observes that one location is 55.2 feet above sea level and another is 46.8 feet below sea level.

1. What rational number represents a location of 55.2 feet above sea level?

2. What rational number represents a location of 46.8 feet below sea level?

3. Plot and label each rational number on the number line.



4. What does zero represent in this situation?

Unit 3, Lesson 6: Comparing a Quantity and Its Opposite in Real-World Context



Benchmark

MA.6.NSO.1.2 Given a mathematical or real-world context, represent quantities that have opposite direction using rational numbers. Compare them on a number line and explain the meaning of zero within its context.



Learning Targets

- ☐ I can compare a quantity and its opposite on a horizontal or vertical line and explain the meaning of zero within the context.
- ☐ I can use the structure of a horizontal and vertical number line to plot rational numbers from a real-world context.



3.6.1 Warm-Up: Ordering Elevations

Four images from locations across the United States are shown.

1. Write an integer that represents each elevation.

a.



Integer: _____

b.



Integer: _____

c.



Integer: _____

d.



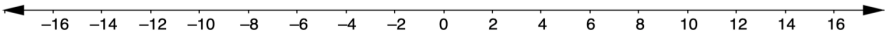
Integer: _____

2. Order the letter of the pictures from lowest elevation to highest.



3.6.2 Collaboration: A Quantity and Its Opposite

- 1.
- a. Plot the number 8 and its opposite on the number line.



- b. What do you notice about the distance of 8 and its opposite from zero?

2. Complete the table for each situation given.

	78.3 Feet Above Sea Level	32°C Below Zero	Withdrawal of \$27.68
Rational Number			
Opposite			
Meaning of Zero			



3. Omar is at the bookstore on Florida Avenue buying a book about training dogs. He also plans to visit Nolan's Pet Shop and Rover's Pet Shop to compare prices for a new leash for his dog. All three stores are on the same street. Nolan's Pet Shop is 7 blocks west of the bookstore. Rover's Pet Shop is 7 blocks east of the bookstore.

- a. Model the locations of the three stores in relation to each other on the number line.

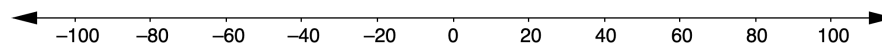
West ← ————— → East

- b. What does zero represent in this situation?
- c. Which pet store is farther away from the bookstore? Explain how you know.

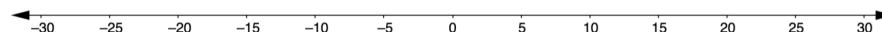


3.6.3 Try It: A Quantity and Its Opposite

1. On the number line, plot and label a deposit of \$64.87 and a withdrawal for the same amount from a bank account. What does 0 represent in this situation?



2. On the number line, locate and label 17°C below zero and 17°C above zero. What does 0 represent in this situation?



3. Carlos is at the ice cream shop. His house is 9.5 blocks north of the shop. The park is 9.5 blocks south of the ice cream shop.
- Plot and label all three places on the number line.
 - When at the ice cream shop, is Carlos closer to the park or his house?



4. Write the rational number that represents the description, the rational number's opposite, and the meaning of the opposite in context of the description.

Description	Rational Number	Opposite	Meaning of Opposite
An atom's positive charge of 7			
A deposit of \$35.84			
3,540 feet below sea level			
A rise of 41.6°C			
A loss of 13.3 pounds			

5. With a partner, choose one description and describe what 0 could represent in that situation.



3.6.4 Bring It Together: A Quantity and Its Opposite

Work with a partner to write answers to the following questions.

1. What is the relationship between any number and its opposite when plotted on a number line?
2. How can you use this relationship to locate the opposite of a given number on the number line?



3.6.5 Your Turn!

1. Janney notices that her room's temperature is 4°F below the recommended temperature, and Ivan's bedroom temperature is 4°F above the recommended temperature. Plot the values on the number line and explain what zero represents in this situation.



2. Finn wants to visit his friend Simon who lives 22.7 blocks west. He also wants to visit his grandparents who live 22.7 blocks east.

- a. Plot and label the locations of Simon and Finn’s grandparents on the number line.



- b. What does zero represent in this situation?

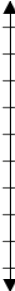
- c. Does Finn live closer to his grandparents or Simon?



3.6.6 Wrap-Up: Ticket Out the Door

Complete the following in the table.

- Write the rational number that represents the situation.
- Write the opposite of the rational number.
- Represent the rational number and its opposite on the number line with appropriate labels.
- Explain the meaning of zero in the situation.

	A Deposit of \$9.34	17°C Below Zero
Rational Number		
Opposite		
Number Line		
Meaning of Zero		



Unit 3, Lesson 7: Absolute Value



Benchmark

MA.6.NSO.1.3 Given a mathematical or real-world context, interpret the absolute value of a number as the distance from zero on a number line. Find the absolute value of rational numbers.



Learning Targets

- ☐ I can find the absolute value of a rational number.
- ☐ I can find the absolute value of a value and interpret it within the context.



3.7.1 Warm-Up: Would You Rather?

1. Would you rather have Pile A or Pile B? Justify your reasoning.

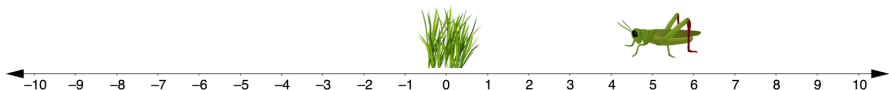


Pile A	Pile B
14 pennies	2 pennies
7 nickels	4 nickels
2 dimes	2 dimes
1 quarter	2 quarters



3.7.2 Exploration: Jumping Bugs

A bug is jumping around on the number line. His home is located at 0. Use a cut out of the bug to answer the following questions about the bug's location.



1. The bug starts at 5 and jumps 3 units to the right.
 - a. What is the bug's new location?
 - b. How far away from home is the bug?

2.
 - a. If the bug starts at 2 and jumps 5 units to the left, what is the bug's new location?
 - b. How far away from home is the bug?

3. Mr. Rodriguez told his class to begin with the bug at home. Then the bug should jump 4 units. When Micah and Brianna did this, they discovered that their bugs were in two different locations. Discuss with your partner how this may have happened. Write your explanation or draw a picture of what could have happened.



3.7.3 Guided Instruction: Absolute Value and Zero

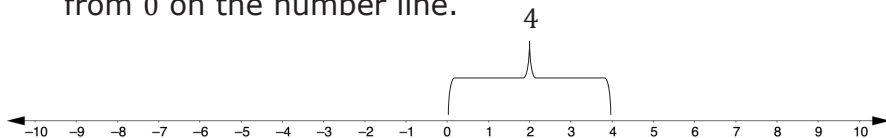
Recall how opposite numbers relate to zero. Complete the explanation below by writing a word in each blank.

1. Pairs of opposite values are the same _____ from zero on the number line. They are mirror images of each other, with a line of symmetry through _____.

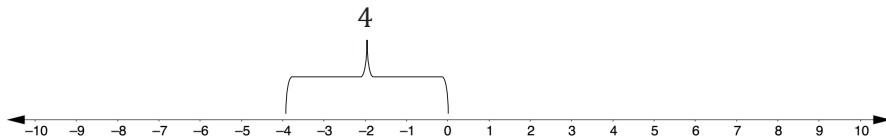


Absolute value is the distance a number is from zero (0) on a number line.

2. The absolute value of 4 is _____ because it is 4 units from 0 on the number line.



3. The absolute value of -4 is _____ because it is 4 units from 0 on the number line.



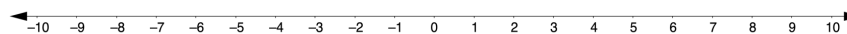
4. The values 4 and -4 are opposites. Each value is 4 units from zero. A number and its opposite have the same _____.



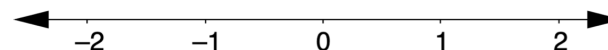
3.7.4 Guided Practice: Absolute Value and Zero

Absolute value is represented using the notation $|a|$. The notation $|a|$ is read, "the absolute value of a ."

1.
 - a. The $|-8| = \underline{\hspace{1cm}}$ because the distance from -8 to 0 is _____.
 - b. What other number has the same absolute value as -8 ?
 - c. Use the number line to show the other number that has the same absolute value as -8 .



2. On the number line, plot and label all numbers with an absolute value of $\frac{3}{2}$.



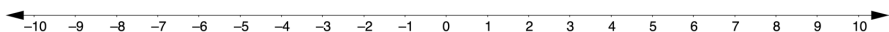
3. Hector and his partner, Salih, were given the following two problems to complete.

$$|5.5| = \underline{\hspace{2cm}} \qquad |\underline{\hspace{2cm}}| = 3.2$$

Hector and Salih had the same answer for one of the blanks but two different answers for the other. Their teacher told them they both had the correct answers.

- a. Which problem can have more than one correct answer?

- b. Use the number line to explain how this is possible.

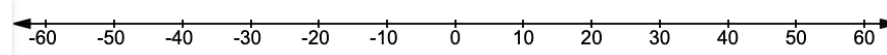


3.7.5 Collaboration: Absolute Value and Opposites in the Real World

1. A tiger was rescued by The Brevard Zoo in Melbourne, Florida. The zoo veterinarian examined the tiger and determined the animal's weight to be 29.6 pounds below average.

- a. What value represents this situation?
- b. What is the absolute value of the value that represents the tiger's weight loss?

- c. Model the tiger's weight loss on the number line.



- d. What does 0 represent in this situation?
- e. If the tiger's weight had been 29.6 pounds above average instead of below, what would be the same about this situation?

2. Simone is doing research on different rides at the Strawberry Festival. She attaches an accelerator to a bumper car to measure the velocity of her car. She recorded the velocities, in miles per hour (mph), in the table.

Speed is the absolute value of velocity.

Time (seconds)	Velocity (mph)	Time (seconds)	Velocity (mph)	Time (seconds)	Velocity (mph)
5	-3.1	20	-2.4	35	-5.2
10	-2.7	25	-6.3	40	3.1
15	4.8	30	2.7	45	-7.8

- Determine when the bumper car is at the same speed. List the times and the speed of the car at those times.
- At what time is the bumper car traveling at the slowest speed?
- At what time is the bumper car traveling at the fastest speed?

3. Before Bryan goes snow skiing, he asks a friend about the weather. The friend says that the temperature is 5°C away from 0.

Select all the temperatures that are possible.

- ☐ -10°C
☐ -5°C
☐ 0°C
☐ 5°C
☐ 10°C

4. A part of the city of New Orleans is 19 feet below sea level. We can use " -19 feet" to describe its elevation, and " $|-19|$ feet" to describe its vertical distance from sea level. In the context of elevation, what do each of the following numbers describe?

Value	Meaning in Context
25 feet	
$ 25 $ feet	
-8 feet	
$ -8 $ feet	

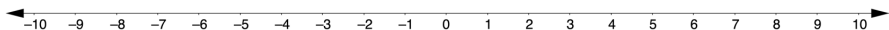




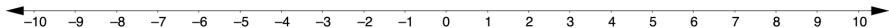
3.7.6 Your Turn!

- Complete each statement and model it on the number line given.

a. $|-7.8| = \underline{\hspace{2cm}}$



b. $|\underline{\hspace{1cm}}| = 3\frac{1}{2}$



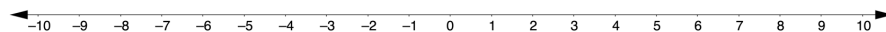
- The absolute value of the elevation of Enterprise, Florida is 19.7 feet. In relation to sea level, write the two values that could represent the elevation of Enterprise, Florida.



3.7.7 Wrap-Up: Ticket Out the Door

- Complete the statement and model it on the number line.

$\left|-\frac{10}{4}\right| = \underline{\hspace{2cm}}$



- The number -220 represents a withdrawal in dollars from a checking account and 220 represents a deposit in dollars to the same checking account. What do the absolute values of these values mean in terms of this context?

Unit 3, Lesson 8: Comparing and Performing Operations With Absolute Value



Benchmark

MA.6.NSO.1.4 Solve mathematical and real-world problems involving absolute value, including the comparison of absolute value.



Learning Targets

- ☐ I can compare the absolute value of two numbers.
- ☐ I can perform operations that involve the absolute values of two numbers.



3.8.1 Warm-Up: Rational Numbers in the Real World

For each of the following problems, ask a classmate for his/her explanation. Once you have received their explanation, write a summary in the space provided in the box. Your classmate should read your summary and initial that it is correct.

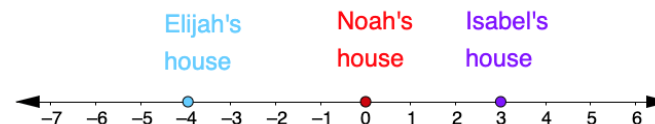
Repeat this with a different classmate for each of the remaining problems.

- | | |
|--|---|
| <p>1. Describe the relationship between two numbers that have the same absolute value.</p> | <p>2. In an American football game, the offensive team makes gains or losses on each play. Gains are represented by positive values and losses by negative values. What does 0 represent in this context?</p> |
|--|---|

Initials: _____

Initials: _____

3. Elijah, Noah, and Isabella live on the same street. The locations of their houses are modeled on the number line.



Describe where they live relative to each other in two sentences. Include the words distance and absolute value.

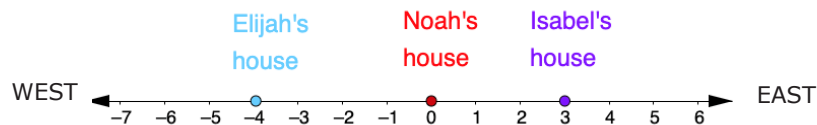
Initials: _____



3.8.2 Exploration: Comparing and Interpreting Absolute Value in Context

- Recall this scenario from the Warm-Up with some additional information.

Elijah, Noah, and Isabella live on the same street that runs west to east. The locations of their houses are modeled on the number line, where each unit represents 1 kilometer.



- Noah's house is located at 0 on the number line model. Represent the distance from Noah's house to both Isabella's house and Elijah's house using absolute value.

	Elijah's House	Isabella's House
Distance from Noah's House	$ \underline{\quad} $	$ \underline{\quad} $

- How do the absolute values in the table compare to each other?

$|\underline{\quad}| > |\underline{\quad}|$

- Isabella lives $\underline{\quad}$ kilometers to the

☐ east
☐ west

 of Noah. Elijah lives $\underline{\quad}$ kilometers to the

☐ east
☐ west

 of Noah. Isabella lives

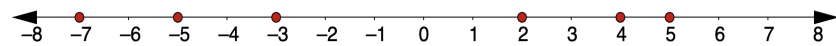
☐ closer to
☐ farther from

 Noah than Elijah does.



3.8.3 Guided Instruction: Comparing Absolute Values

A number line is shown with the integers $-7, -5, -3, 2, 4,$ and 5 plotted.



- Discuss with your partner which statements are true. Be prepared to explain your thinking.

Mathematical Statement	Explanation
$-7 < -3$	
$ -7 $ is 7	
$ -3 $ is 3	
$ -7 < -3 $	

- Does the comparison of -7 and -3 match the comparison of $|-7|$ and $|-3|$?

3. With your partner, discuss why comparing two negative values and comparing the absolute value of the same two negative numbers may be different. Summarize your discussion below.

4. For each row, use the symbols $>$, $<$, or $=$ to compare each pair of values.

	Inequality	Absolute Value Inequality
Pair 1	$4 \underline{\hspace{1cm}} -5$	$ 4 \underline{\hspace{1cm}} -5 $
Pair 2	$-3 \underline{\hspace{1cm}} 2$	$ -3 \underline{\hspace{1cm}} 2 $
Pair 3	$6 \underline{\hspace{1cm}} -6$	$ 6 \underline{\hspace{1cm}} -6 $

5. What do you notice about the comparisons?

6. When comparing the absolute values of two numbers, numbers that are closer to zero will always have the

_____ absolute value.



3.8.4 Collaboration: Inequality Mix and Match

Work with your partner to complete the table.

Partner A will select two numbers and one comparison symbol. At least one of the numbers must be written as an absolute value.

Partner B will then write the mathematical statement in words. The following phrases may be helpful: is equal to, the absolute value of, is greater than, is less than.

Switch roles until each partner has taken two turns. An example is shown in the first row.

Numbers					Symbol
-0.7	$ -9 $	11	14	$ -8 $	$<$
$-\frac{6}{2}$	-2.2	$ 2.5 $	8	$ 0.7 $	$=$
$ -14 $	$ 20 $	$\frac{7}{2}$	$ 15 $	$ \frac{5}{2} $	$>$

Partner	Mathematical Statement	Statement in Words
Example	$ 3 < 4$	<i>the absolute value of 3 is less than 4</i>
Partner A		
Partner B		
Partner A		
Partner B		

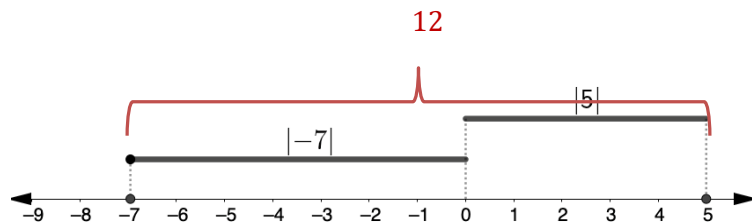




3.8.5 Guided Instruction: Operations Using Absolute Value

1. Ty'shaun and Gabby were asked to explain their strategies for determining the sum of $|-7|$ and $|5|$.

Ty'shaun modeled his thinking on the number line.



Gabby's explanation was "I know $|-7|$ is a distance of 7 from 0 and $|5|$ is a distance of 5 from 0. The sum of 7 and 5 is 12, so the sum of $|-7|$ and $|5|$ is also 12."

Explain which strategy makes more sense to you.

2. Find the value of each expression.

a. $|-18| - |-13|$

b. $12 \times |-4|$

c. $\frac{|45|}{|-3|}$

d. $|-22| + |31| + |40|$



3.8.6 Your Turn!

1. Use the symbols $>$, $<$, or $=$ to compare each pair of values.

a. $|-42|$ ____ -25

d. $|-512|$ ____ 284

b. -21 ____ $|18|$

e. $|-176|$ ____ $|176|$

c. $|-582|$ ____ $|-790|$

f. $|-38|$ ____ 0

2. An octopus is 47 feet below sea level at the same time a dolphin is at 108 feet below sea level.

- a. Write a value to describe the position of each sea animal relative to sea level.

Octopus	Dolphin

- b. Use an inequality symbol, $<$ or $>$, to write an inequality comparing the distance each sea animal is from sea level.

| ____ | ____ | ____ |

- c. What does this inequality statement indicate about the locations of each sea animal?

3. Find the value of each expression.

a. $|-58| + 12 =$ ____

b. $|-13| \times 5 =$ ____

c. $\frac{|-36|}{|9|} =$ ____



3.8.7 Wrap-Up: Ticket Out the Door

1. The table shows the elevation, in meters (m), of five locations.

Location	Elevation (m)
Vpadina Kaundy, Kazakhstan	-132
Laguna del Carbón, Argentina	-105
Zuidplaspolder, Netherlands	-7
Fort Myers, Florida	2
Britton Hill, Lakewood, Florida	105

For each pair, use $<$, $>$, or $=$ to compare their distances from sea level.

Zuidplaspolder, Netherlands	____ ____ ____
Fort Myers, Florida	

Britton Hill, Lakewood, Florida	____ ____ ____
Laguna del Carbón, Argentina	

Laguna del Carbón, Argentina	____ ____ ____
Vpadina Kaundy, Kazakhstan	

Unit 3, Lesson 9: Operations With Absolute Value in the Real World



Benchmark

MA.6.NSO.1.4 Solve mathematical and real-world problems involving absolute value, including the comparison of absolute values.



Learning Target

- ☐ I can perform operations that involve the absolute value of two numbers.



3.9.1 Warm-Up: Math Fail

1. What do you notice about the sign?



3.9.2 Collaboration: Absolute Value and Profit

1. Landon starts a lawn-mowing business to earn money. All of his profits are being saved for college.

Profit is the difference between the amount earned (revenue) and the amount spent.

His father lends him a lawnmower to use for free, but he must pay for the gas and other supplies. For each lawn, his expenses are \$7.30, and he earns an average of \$40.00.

- a. In this situation, revenue is represented with a ☐ positive ☐ negative value and an expense is represented

by a ☐ positive ☐ negative value.

- b. Write a value to represent each situation.

Landon's Finances	Value
Expenses per lawn	
Revenue per lawn	

c. Use an inequality symbol, $<$ or $>$, to compare the absolute values of Landon’s expenses to revenue for each lawn he mows.

$$| \quad | > | \quad |$$

d. How does the inequality show that Landon will make a profit?

e. To make a profit, total expenses of a business must be

☐ greater than
 ☐ less than

 total revenue. A business is not profitable when the total

☐ expenses are
 ☐ revenue is

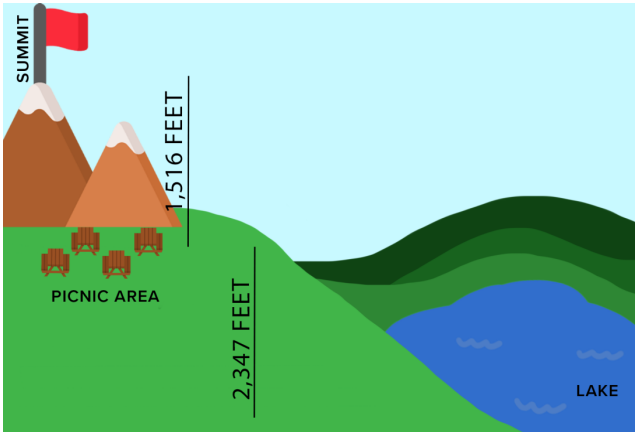
 greater than the

☐ expenses.
 ☐ revenue.



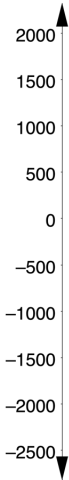
3.9.3 Guided Instruction: Operations With Absolute Value in the Real World

1. Javier and his family drove to a picnic area on a mountain. From the picnic area, the family hiked to the mountain’s summit, which was an elevation change of 1,516 feet (ft). They hiked back to the picnic area to eat lunch. After lunch, they hiked down to the lake, which was an elevation change of 2,347 ft.



a. In the table, write the value that represents the position of each location if the value of the position of the picnic area is 0. Then, plot each value at the estimated location on the number line.

Location	Value
Picnic area	0
Mountain summit	
Lake	



- b. Represent the distance of each location from the picnic area.

Location	Distance from Picnic Area
Mountain summit	
Lake	

- c. What operation would you use with the representations in the table to find the distance from the lake to the mountain summit?
- d. What is the distance from the lake to the mountain summit?
- e. What operation would you use to determine how much farther a hike to the lake is than a hike to the mountain summit?
- f. How much farther is a hike to the lake than a hike to the mountain summit?

2. Chloe is starting a painting business. The supplies she purchased for the business cost \$825. She will earn a fee of \$80 for each room that she paints. Chloe wants to determine how many rooms she will need to paint before she starts making a profit.

- a. ☐ Fees Chloe earns increase her profit.
☐ Supplies Chloe buys
- b. ☐ Fees Chloe earns decrease her profit.
☐ Supplies Chloe buys
- c. Write a value in the blank for each absolute value and a word in the remaining blank to complete each sentence.

To find the revenue, Chloe multiplies the number of rooms she paints by |____|.

To find the profit, Chloe subtracts |____| from the _____.

- d. In the table, complete the mathematical statement and then perform the operations to determine if the number of rooms painted would result in a profit.

Number of Rooms	Mathematical Statement	Profit (Yes or No)
6	6 ____ — ____	
10	10 ____ — ____	
13	13 ____ — ____	
16	16 ____ — ____	



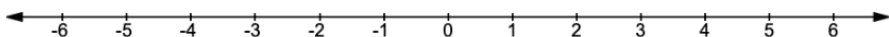


3.9.4 Try It: Operations With Absolute Value in the Real World

Logan, Mia, and Xavier all live on the same street as their school, which runs from east to west.

- Logan lives 4 blocks to the west of the school.
- Mia lives 5 blocks to the east of the school.
- Xavier lives 2 blocks to the west of the school.

1. Use the number line to model the position of each house and the school along the street. Label the location of each person's house and the school.



2. Work with your partner to write the mathematical sentence using absolute values. Then, use the mathematical sentence to determine each answer.
 - a. How far does Logan live from Mia?
 - b. How far does Mia live from Xavier?
 - c. How far does Xavier live from Logan?
 - d. Was the same operation used to find each distance?



3.9.5 Collaboration: Halfway Point

Mount Ojos del Salado is the highest mountain in Chile. Its peak is 6,893 meters above sea level. The lowest point of the Atacama Trench, just off the coast of Peru and Chile, is 8,065 meters below sea level.

Work with your partner to answer each question.

1. Determine how to label the number line so that you can approximate a location on the number line that represents Mount Ojos del Salado, Atacama Trench, and sea level.
2. Label the number line and place a point at the approximate location for Mount Ojos del Salado, Atacama Trench, and sea level.
3. Write a mathematical sentence using absolute value to find the difference in elevations between Mount Ojos del Salado and the Atacama Trench.



4. With your partner, discuss how the halfway point between the peak of Mount Ojos del Salado and the Atacama Trench can be found. Write down all of the strategies that were discussed.

5. Explain how to determine if the halfway point's elevation is above sea level or below sea level.



3.9.6 Your Turn!

1. The captain of a fishing vessel is standing on the deck at 23 feet above sea level. He holds a rope tied to a fishing net below him that is underwater at a depth of 38 feet.
- a. Draw a model of this situation using the number line.
- b. Write a mathematical sentence using absolute value to determine the length of the rope.



2. Aryan owes \$325.46 for recent purchases he made using his credit card, and he owes his parents another \$187.27.
- a. Write a mathematical statement using absolute value that can be used to determine Aryan's total debt.
- b. What is the value of Aryan's total debt?
3. When Brenda places a package of chicken in the freezer, the chicken is at a temperature of 41° Fahrenheit ($^{\circ}\text{F}$). While in the freezer, the temperature of the chicken changes -2°F every hour until it reaches the temperature of the freezer. What is the temperature of the chicken 3 hours after Brenda puts it in the freezer?



3.9.7 Wrap-Up: Reflection

1. Check all of the statements below that describe your understanding of and ability to work with absolute values.
- ☐ I can compare the absolute value of two numbers.
 - ☐ I need support on how to compare the absolute value of two numbers.
 - ☐ I can perform operations that involve the absolute value of two numbers.
 - ☐ I need support to perform operations that involve absolute value.
 - ☐ I can interpret comparisons of absolute values in real-world contexts.
 - ☐ I need support interpreting comparisons of absolute values in real-world contexts.
 - ☐ I can perform operations that involve absolute value when solving real-world problems.
 - ☐ I need support to perform operations that involve absolute value when solving real-world problems.